

6.6 Graphing Two-Variable Linear Inequalities for a Real-World Context

Lesson Introduction

 Benchmarks	MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.		MA.912.AR.9.6 Given a real-world context, represent constraints as a system of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.		 Learning Targets	- I can graph the solution set of a real-world linear inequality, given in standard form, slope-intercept form, or point-slope form on a coordinate plane. - I can relate the key features of the inequality to the context. - I can interpret solutions to problems as viable or non-viable options.
 Benchmark Coherence	Building On			Working Toward		
	N/A			MA.912.AR.3.10 Given a mathematical or real-world context, graph the solution set to a two-variable, quadratic inequality.		
 Essential Questions	- How do you graph the solution set of a real-world, linear inequality? - How do you interpret solutions to a real-world, linear inequality as viable or non-viable?					
 Lesson Narrative	This lesson extends students' learning about graphing two-variable, linear inequalities from mathematical contexts to real-world contexts. Students will learn how to represent real-world contexts using two-variable inequalities as well as graphing them, solving them, and relating solutions to the context.					
 Component Summary	6.6.1 Warm-Up (~5 min)	Hebi puzzle activity (<i>SE p. 324</i>)		6.6.4 Guided Practice (10 min)	Practice relating two-variable inequalities and their solutions to real-world contexts (<i>SE p. 327</i>)	
	6.6.2 Collaboration (5-10 min)	U-Pick berries real-world collaboration activity (<i>SE p. 325</i>)		6.6.5 Try It (10 min)	Practice relating two-variable inequalities and their solutions to real-world contexts (<i>SE p. 328</i>)	
	6.6.3 Guided Instruction (10 min)	Relate two-variable inequalities and their solutions to real-world contexts (<i>SE p. 325</i>)		6.6.6 Wrap-Up (~5 min)	Reflect on understanding of lesson learning targets (<i>SE p. 329</i>)	
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A		(Optional) Consider preparing examples of stocks to show students during 6.6.4 Guided Practice.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse when to use a dotted or solid line when graphing the inequality. - Students may shade in the incorrect region when graphing an inequality.	This lesson is comprised primarily of word problems - consider your struggling readers when rolling out this lesson. Leverage any existing strategies throughout the lesson to support students in real-world word problems, including utilizing either the Three Reads or Mark-Up routines.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson consists completely of word problems. Consider the use of the Three Reads routine to help students navigate these types of problems. This routine supports reading comprehension of problems and meta-awareness of mathematical language in context.	MA.K12.MTR.7.1: Real-World Contexts During this lesson, students will connect mathematical concepts to a variety of everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.
 Differentiate Instruction	Consider going digital with this lesson. The questions in the components of this lesson are scaffolded to bring students into the problem and connect its content with their mathematical knowledge. Leverage an individual response digital tool such as Nearpod, Pair Deck, etc. where each sub-question can be responded to by students to see who understands each piece before moving on. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

6.6.1 Warm-Up: Hebi Puzzle

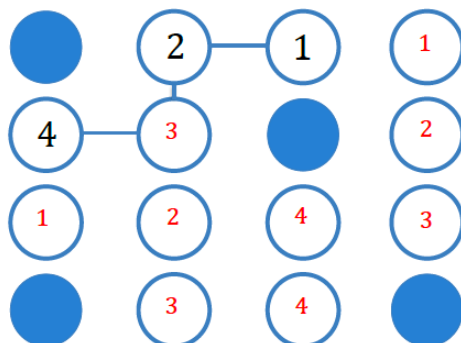
Component Overview: Students will complete a Hebi Puzzle. The purpose of this Warm-Up is to engage students in mathematical problem solving with a low entry point.



6.6.1 Warm-Up: Hebi Puzzle

Hebi is a Japanese number puzzle. The word Hebi means snake. The goal is to place number snakes on the grid so that each snake connects the digits with consecutive numbers joining vertically or horizontally to adjacent circles. Numbers never connect diagonally. Snakes cannot cross blue circles.

1. Use the numbers 1 – 4 to fill in the empty circles so that a string of numbers can be connected in numerical order.



Consider showing an example of moves students can and cannot make based on the description on the game before releasing students to try to solve the puzzle, if needed.

6.6.2 Collaboration: U-Pick Berries

Component Overview: Students will collaborate on interpreting the key information from the U-Pick Berries context. For this activity, students will need a partner.



6.6.2 Collaboration: U-Pick Berries

5K Farms is an organic berry farm in Plant City, Florida. In the spring, 5K Farms has U-Pick strawberries for \$2 per pound and blackberries for \$5 per pound. Aaron is going berry picking and has \$20.

1. Determine whether each of the following scenarios is possible given the information above.

Scenario	Cost	Is this possible?
Aaron picks 3.2 pounds of blackberries and 1.9 pounds of strawberries.	\$19.80	☐ Yes ☐ No
Aaron picks 2.7 pounds of blackberries and 3.8 pounds of strawberries.	\$21.10	☐ Yes ☐ No
Aaron picks just 4.5 pounds of strawberries and no blackberries.	\$9.00	☐ Yes ☐ No

2. Compare your responses with your partner and resolve any differences.

Answers will vary.



The purpose of this activity is to get students to explore this context to decipher the information and draw reasonable conclusions using the table of questions on this page.

At the end of this activity, have students share their answers out to question 2 to provide context for the next component.

6.6.3 Guided Instruction: Connecting Real-World Contexts and Two-Variable Linear Inequalities

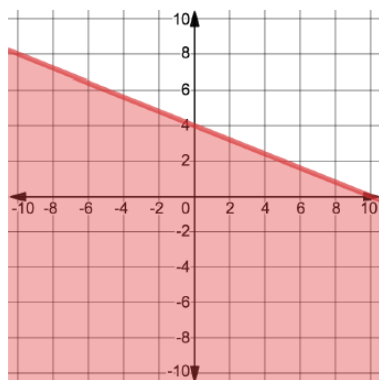
Component Overview: Students will learn how to relate two-variable, linear inequalities to real-world contexts using the context from the previous component.



6.6.3 Guided Instruction: Connecting Real-World Contexts and Two-Variable Linear Inequalities

Aaron's scenario can be represented by a linear inequality in two variables.

1. To determine the combinations of each type of berry he could buy, Aaron wrote the inequality $2x + 5y \leq 20$.
 - a. Define the variables Aaron used based on the context.
 Let x represent the number of pounds of strawberries Aaron buys.
 Let y represent the number of pounds of blackberries Aaron buys.
 - b. Explain why the inequality sign $\leq$ makes sense in this context.
Answers will vary. The inequality symbol $\leq$ makes sense in this context because he can spend at most \$20.
 - c. Graph the inequality $2x + 5y \leq 20$.



In the next components, students will have an opportunity to engage with different real-world contexts and relate inequalities and solutions to those contexts. The intention of this component is for the teacher to model strategies for students to use on their own. This problem should be guided and include think-aloud and mark-up strategies that would help students on their own in future problems.



For question 1b, “less than” makes sense because he can’t spend more than \$20 since that’s all he has, and “or equal to” makes sense because it’s possible for him to spend exactly \$20.

6.6.3 Guided Instruction: Connecting Real-World Contexts and Two-Variable Linear Inequalities (continued)

- d. The ordered pair $(-1, 2)$ is a solution to the inequality $2x + 5y \leq 20$. Explain why this is not a viable solution in the context of buying berries.

Answers will vary. The ordered pair $(-1, 2)$ is not a viable solution in the context because it would mean that Aaron bought 2 pounds of blackberries and -1 pounds of strawberries, and it's not possible to buy a negative number of pounds of strawberries.

- e. With your partner, discuss which solutions to the mathematical inequality are not viable solutions in this context.

Answers will vary. $(-5, 6)$ would not be viable because he cannot purchase negative amounts.

- f. Select the additional constraints necessary on x and y in order for a solution to be viable in the berry-picking context.

- ☐ $x > 0$
- ☒ $x \geq 0$
- ☐ $y > 0$
- ☒ $y \geq 0$

- g. The ordered pair $(4.3, 2.28)$ is a point on the boundary line. Substitute the ordered pair into the inequality $2x + 5y \leq 20$.

$$\begin{aligned} 2(4.3) + 5(2.28) &\leq 20 \\ 8.6 + 11.4 &\leq 20 \\ 20 &\leq 20 \end{aligned}$$

- h. Explain what ordered pairs on the boundary line represent in the situation.

Answers will vary. If an ordered pair is on the boundary line, it means that Aaron would pay exactly \$20 for that combination of berries.



Students should understand that any solutions with one or both variables negative would not be viable solutions in this real-world context.



Both should be greater than or equal to 0. Point to the third scenario from 6.6.2 Collaboration if students do not understand why it is possible to be equal to.

6.6.4 Guided Practice: Two-Variable Linear Inequalities in Real-World Contexts

Component Overview: Students will engage in Guided Practice with real-world, two-variable, linear inequalities using one scenario with multiple sub-questions.



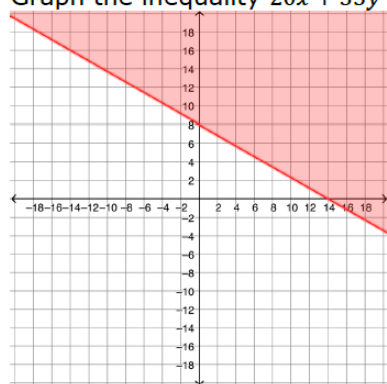
6.6.4 Guided Practice: Two-Variable Linear Inequalities in Real-World Contexts

1. Jolene invested some of her savings in the stock market. She bought 20 shares of a tech company and 35 shares of an automotive company. She hopes to make a combined profit of at least \$280 when she sells her shares. She wrote the inequality $20x + 35y \geq 280$ to represent the situation, where x is the change in value of one share of the tech company and y is the change in value of one share of the automotive company.

- a. Explain what it would mean if x or y were a negative number.

Answers will vary. If x or y were negative, it would mean that the value of the stock has decreased since Jolene bought it.

- b. Graph the inequality $20x + 35y \geq 280$.



- c. Explain what the point $(-2.25, 15.75)$ represents in Jolene's situation.

Answers will vary. In this situation, the price of a share of the tech company would have dropped by \$2.25, but the price of a share of the automotive company would have increased by \$15.75, and Jolene would still make more than \$280.



Allow informal observation throughout the previous two components to dictate how “guided” this Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



For question 1a, have students share what they think this means. Ensure the language is correctly used with the context in which it is provided.



In question 1c, how can the point be a solution but also be a negative value? What does that mean in this scenario?

6.6.4 Guided Practice: Two-Variable Linear Inequalities in Real-World Contexts (continued)

- d. Explain why it makes sense in Jolene's context that there are no solutions for which both x and y are both negative.

Answers will vary. It makes sense because if Jolene loses money on one of the stocks, she has to make money on the other stock in order for her total to be at least \$280.

Stock prices in the United States are rounded to the hundredths place.

- e. Complete the statement.

In order for solutions to the mathematical inequality to be viable solutions in the real-world context, the x and y values must be

- ☐ greater than 0.
- ☐ non-negative.
- ☐ integers.
- ☒ rounded to 2 decimal places.

- f. Explain what a solution on the boundary line of the inequality would mean in the real-world context.

Answers will vary. If a solution was on the boundary line, it would mean that Jolene would make a profit of exactly \$280 if she sold her stocks.



Consider increasing engagement in this scenario by showing students real-world stocks that students can connect to.



Allow students to share out their answers to question 1f as a whole-group discussion opportunity.

6.6.5 Try It: Representing Real-World Contexts With Two-Variable Inequalities

Component Overview: Students will try a real-world application problem with a partner.

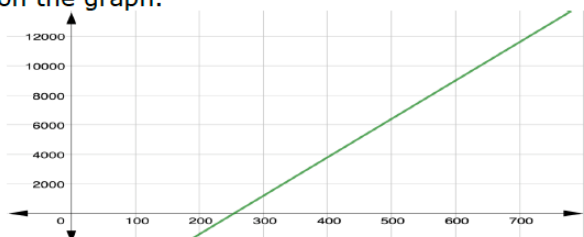


6.6.5 Try It: Representing Real-World Contexts with Two-Variable Inequalities

1. In the 2018-2019 harvest year, Florida produced approximately 43.7% of the total citrus produced in the United States. Given the acreage of land and data collected by the Florida Department of Agriculture and Consumer Services, citrus production can be predicted by the inequality $y \leq 26x - 6,600$, where y is Florida's annual production of citrus in thousands of tons and x is the number of thousands of acres of land.
 - a. Select the statements that could be reasons why the inequality modeling the relationship between acreage and citrus production uses the symbol $\leq$.
 - ☐ The acreage of land that actually bears fruit is less than or equal to the total acreage.
 - ☐ Extreme weather and/or disease could result in a less abundant citrus crop that year.
 - ☐ Citrus trees may start to produce more as they get older.
 - b. Discuss the statements that you chose with your partner.

Answers will vary.

The boundary line of the inequality, $y = 26x - 6,600$, is shown on the graph.



- c. Discuss with your partner which region to shade to represent the inequality $y \leq 26x - 6,600$.

Answers will vary.



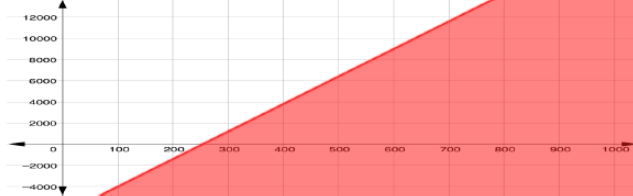
Have students discuss these statements in question 1a, including how the statements connect to the real-world problem. Consider prompting students to mark up the text as they complete and discuss questions 1a - 1c.



How can you determine which region would be shaded?

6.6.5 Try It: Representing Real-World Contexts With Two-Variable Inequalities (continued)

- d. Shade the graph of the inequality $y \leq 26x - 6,600$.



- e. The ordered pair $(400, 1,000)$ is in the solution set of the inequality. Explain what this solution represents in the real-world context of citrus production.

Answers will vary. This ordered pair would represent the possible situation in which there are 400,000 acres of land that produce 1,000,000 tons of citrus one year.

- f. The ordered pairs below are all in the solution set of the mathematical inequality. Identify which ordered pair is not a viable solution in the real-world context.
- ☐ $(300, 0)$
 - ☐ $(300, 1,000)$
 - ☒ $(300, -1,000)$

Explain your selection.

Answers will vary. $(300, -1,000)$ would not be a viable solution in the context of citrus production because it would mean that there was a negative number of tons of citrus produced, which is not possible.

- g. Which constraint on the values of x and y makes the solutions viable?
- ☐ Both x and y must be greater than 0.
 - ☒ Both x and y must be non-negative.
 - ☐ Both x and y must be integers.
 - ☐ Both x and y must be rounded to two decimal places.



At the conclusion of this activity, engage the class in a separate discussion about their answers to questions 1e, 1f, and 1g. Allow students to share out their answers and poll the class on agree/disagree to come to a shared understanding of the answer for each part.

6.6.6 Wrap-Up: Reflection

Component Overview: Students will complete a rating of their confidence with each learning target in the lesson.



6.6.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate how you feel about each learning target.

Answers will vary.

- a. I can graph the solution set of a real-world linear inequality, given in standard form, slope-intercept form, or point-slope form on a coordinate plane.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



- b. I can interpret solutions to problems as viable or non-viable options.

I need help.	I'm getting there, but	I understand this
I can't get started.	I need more practice.	and I feel confident.



Consider using student responses from prior work and self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback about students' learning progression.